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Printer

Abstract

An embodiment of the present invention is a printer including: a feed roller configured to feed a print medium; a print head configured to pinch the print medium with the feed roller and to print on the print medium; at least a biasing member configured to bias the print head in a first direction toward the feed roller; and an abutting part that abuts on the print head. The print head comprises: a first side biased by the biasing member, and a second side that is an opposite of the first side, the second side facing the feed roller. When seen from a side view of the printer, a point of applying a biasing force of the biasing member to the print head is between a position at which the print head receives a reaction force from the feed roller and a position at which the abutting part supports the second side of the print head. The print head is swingable around a fulcrum at the abutting part in a clockwise swinging direction and a counterclockwise swinging direction in a side view of the printer.

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Background/Summary

FIELD

(1) The present invention relates to a printer.

BACKGROUND

(2) Methods are known for mounting a thermal head, for example, by fixing the thermal head using screws, a shaft, or a bracket, etc. to an internal frame or a housing, in a conventional thermal printer having a thermal printer.

(3) Japanese Unexamined Patent Application Publication No. 2019-064017, for example, a thermal printer is disclosed which includes a support member for fixing a thermal head with a support member made of a metal plate. The support member has two elongated plate-shaped arms and a connecting part for connecting the two arms, and the thermal head is mounted on a bottom surface of the connecting part. The two arms are axially supported at distal ends thereof by a side wall of a frame.

BRIEF SUMMARY

Technical Problem

(4) However, the conventional mounting method noted above has a drawback that uniform pressure of the thermal head that abuts on a platen roller may not be obtained, over an axial direction of the platen roller. Print quality may be degraded when uniform pressure of the thermal head that abuts on the platen roller is not obtained.

(5) In view of the above, an object of the present invention is to prevent print quality from being degraded due to a method for mounting a thermal head.

Solution to Problem

(6) An embodiment of the present invention is a printer including: a feed roller configured to feed a print medium; a print head configured to pinch the print medium with the feed roller and to print on the print medium; at least a biasing member configured to bias the print head in a first direction toward the feed roller; and an abutting part that abuts on the print head. The print head comprises: a first side biased by the biasing member, and a second side that is an opposite of the first side, the second side facing the feed roller. When seen from a side view of the printer, a point of applying a biasing force of the biasing member to the print head is between a position at which the print head receives a reaction force from the feed roller and a position at which the abutting part supports the second side of the print head. The print head is swingable around a fulcrum at the abutting part in a clockwise swinging direction and a counterclockwise swinging direction in a side view of the printer.

Advantageous Effects

(7) The embodiment of the present invention prevents print quality from being degraded due to a method for mounting a thermal head.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1**A is a perspective view of a printer according to an embodiment, in which a printer cover is in a closed state.

(2) FIG. **1**B is a perspective view of the printer of the embodiment, in which the printer cover is in an open state.

(3) FIG. **2** is a perspective view of the printer of the embodiment, in which the printer cover is in the open state, a peeling unit is in an open state, and a paper roll is not contained.

(4) FIG. **3** shows partial sectional views for explaining continuous issuing and peeling issuing in the printer of the embodiment.

(5) FIG. **4** illustrates a mechanism for making the printer cover be in the open state by using a cover open button.

(6) FIG. **5** shows positional relationships between a platen-holding bracket and levers.

(7) FIG. **6** shows perspective views of the peeling unit when open and when closed.

(8) FIG. **7** is a perspective view of the peeling unit when open, as seen from a viewpoint different from that of FIG. **6**.
(9) FIG. **8** illustrates relationships between a peeling unit open lever and the peeling unit when continuous issuing is performed and when a peeling unit open button is operated.
(10) FIG. **9** illustrates relationships between the peeling unit open lever and the peeling unit when continuous issuing is performed and when the peeling unit open button is operated.
(11) FIG. **10**A is a plane view of the printer cover of the printer of the embodiment, and FIG. **10**B is an A-A cross section in FIG. **10**A.
(12) FIG. **11** is an enlarged sectional view of a part in the vicinity of a peeling roller when peeling issuing is performed.
(13) FIG. **12** illustrates movements of folding down the peeling unit.
(14) FIG. **13** illustrates movements of folding down the peeling unit.
(15) FIG. **14** sequentially shows movements in switching from continuous issuing to peeling issuing in the printer of the embodiment.
(16) FIG. **15** sequentially shows movements in switching from continuous issuing to peeling issuing in the printer of the embodiment.
(17) FIG. **16**A shows a front side of a thermal head, and FIG. **16**B shows a rear side of the thermal head.
(18) FIG. **17** shows enlarged sectional views of an A-A cross section and a B-B cross section in FIG. **16**A.
(19) FIG. **18** is a partial sectional view of the printer, including a shaft-receiving groove for the thermal head.
(20) FIG. **19** illustrates a method of replacing the thermal head.
(21) FIG. **20** shows a protrusion that is provided to an internal frame so as to support the thermal head.
(22) FIGS. **21**A and **21**B illustrate forces that act on the thermal head in the printer of the embodiment; FIG. **21**A shows a cross section in a plane perpendicular to an upper-lower direction, and FIG. **21**B shows a cross section in a plane perpendicular to a right-left direction.
(23) FIG. **22**A is a perspective front view of a thermal head of another embodiment, and FIG. **22**B is a perspective rear view of the thermal head of the another embodiment.
(24) FIG. **23** is a perspective view of a plate member included in the thermal head of the another embodiment.
(25) FIG. **24** is a perspective view of the thermal head of the another embodiment, as seen from a viewpoint different from that of FIG. **22**.
(26) FIG. **25** is a side view showing a positional relationship between the platen-holding bracket and the thermal head of the another embodiment.

DETAILED DESCRIPTION

Schematic Structure of Printer **1**

(27) A printer **1** according to one embodiment of the present invention is a label printer in which continuous issuing and peeling issuing can be switched. Hereinafter, the printer **1** will be described in detail with reference to the attached drawings.
(28) It is noted that directions of up (UP), down (DN), left (LH), right (RH), front (FR), and rear (RR) are defined in each drawing, for example, as illustrated in the perspective views of FIGS. **1**A and **1**B, but these definitions of directions are made mainly for convenience of explanation of drawings and are not intended to limit an in-use position of the printer of the present invention.
(29) In these definitions of directions, a “printer front-rear direction” means a front-rear direction of the printer **1**. A “printer width direction” means a right-left direction or a lateral direction of the printer **1**.
(30) Each of FIGS. **1**A, **1**B, and **2** is a perspective view of the printer **1** of this embodiment. FIG. **1**A shows a case in which a printer cover **3** is in a closed state. FIGS. **1**B and **2** show cases in which

the printer cover **3** is in an open state. FIG. **1**B shows a state in which a paper roll “R” is set. FIG. **2** illustrates a paper roll “R” and shows a state of the printer **1** before the paper roll “R” is set.
(31) As shown in FIG. **1**A, the printer **1** has a body case **2** and the printer cover **3** that protect internal functional components. The printer **1** has an upper surface provided with an ejection part **20** for ejecting labels.
(32) It is possible to use the printer **1** with the ejection part **20** facing upward (in a horizontally placed state); however, the printer **1** can also be used with the ejection part **20** facing a horizontal direction (in a vertically held state), such as by hanging a belt hook (not shown) provided on a bottom of the printer **1**, on a belt of an operator, or by attaching a shoulder strap (not shown) to the printer **1** and putting it on a shoulder of an operator.
(33) A display panel **15** is provided on a front side of the ejection part **20** in the body case **2**. The display panel **15** may have a touch panel input mechanism for receiving an operation input from an operator. The display panel **15** is connected to a circuit board inside the printer **1** and outputs an image of a user interface related to, for example, an operating state of the printer **1** or operation of the printer **1**, based on a display signal supplied from the circuit board.
(34) Although not shown, an internal frame for supporting or holding various functional components is disposed in the inside of the printer **1**, which is surrounded by the body case **2** and the printer cover **3**. The internal frame, the body case **2**, and the printer cover **3** correspond to a printer body.
(35) The printer cover **3** is able to swing between an open position for exposing the inside of the printer **1** and a closed position for covering the inside of the printer **1**.
(36) In response to operation to a cover open button **51***b* that is provided to the body case **2**, the printer cover **3** opens as shown in FIG. **1**B. Opening the printer cover **3** exposes a paper roll-containing chamber **9**. The paper roll-containing chamber **9** forms space for containing a paper roll “R” (an example of a roll body).
(37) As shown in FIG. **2**, the paper roll “R” has a roll shape into which a strip continuous paper “P” is wound. The continuous paper “P” includes a strip liner PM and a plurality of labels PL that are temporarily attached on the liner PM at predetermined intervals. A label adherend surface of the liner PM is coated with a release agent, such as silicone, in order to easily peel off labels PL. In addition, position detection marks “M” that indicate reference positions of labels PL are formed at predetermined intervals on a back surface of the label adherend surface of the liner PM.
(38) A front side of the label PL is a printing surface to be printed with information, and it is formed with a thermal color developing layer that develops a specific color when reaching a predetermined temperature region. A back side of the printing surface is an adhesive surface coated with an adhesive. The adhesive surface is attached to the label adherend surface of the liner PM, whereby the label PL is temporarily attached on the liner PM.
(39) A pair of paper roll guides **6***a* are placed in the paper roll-containing chamber **9**. The pair of paper roll guides **6***a* are members that rotatably support the paper roll “R” while being in contact with both side surfaces of the paper roll “R” and that guide feeding the continuous paper pulled out of the paper roll “R.” The paper roll guides **6***a* are preferably movable along a width direction of the paper roll “R” in order to vary their positions in accordance with the width of the paper roll “R.”
(40) As shown in FIG. **2**, the printer cover **3** is axially supported to the body case **2** by a hinge **8** so as to swing relative to the body case **2** between the open position and the closed position. The hinge **8** has a hinge shaft **81** that is provided with a torsion spring (not shown) for biasing the printer cover **3** in a direction from the closed position to the open position.
(41) As shown in FIG. **2**, a platen roller **10** (an example of a feed roller) is axially supported in a manner rotatable in forward and reverse directions, at an end of the printer cover **3.** The platen roller **10** is a feeding unit for feeding the continuous paper “P” pulled out of the paper roll “R” and is formed in such a manner as to extend along the width direction of the continuous paper “P.” A

gear **10***b* is coupled to an end of a platen shaft **10***a* of the platen roller **10**. When the printer cover **3** is at the closed position, the gear **10***b* engages with a gear **22***b* that is disposed in the body case **2**, and it is mechanically connected via the gear **22***b* to a roller-driving stepping motor (not shown) or the like.
(42) As shown in FIG. **2**, a peeling bar **12** is placed along and in the vicinity of the platen roller **10**, in the printer cover **3**. The peeling bar **12** is a peeling member for peeling labels PL from the liner PM and is fixed to both side walls of the printer cover **3** at both ends. The peeling bar **12** may be fixed to both ends of the platen shaft **10***a*.
(43) In an embodiment, the cross section of the peeling bar **12** has a substantially triangle shape; however, it is not limited thereto, and it may have a spherical shape or an elliptical shape.
(44) The body case **2** is provided with a platen-holding bracket **27** for holding the platen shaft **10***a* of the platen roller **10** when the printer cover **3** is closed. A thermal head **28** is disposed in front of the platen-holding bracket **27**.
(45) The thermal head **28** (an example of a print head) is a print unit for printing information such as characters, symbols, figures, or bar codes, on labels PL, which are temporarily attached on the liner PM fed out of the paper roll "R." The thermal head **28** is provided so as to face the platen roller **10** when the printer cover **3** is in the closed state.
(46) As described later, a flexible cable that is connected to the circuit board (not shown) is detachably attached to the thermal head **28**. The thermal head **28** includes a plurality of heating elements (heating resistors) that are arranged along the width direction of the continuous paper "P." The thermal head **28** performs printing by selectively energizing the plurality of heating elements based on a signal transmitted from the circuit board.
(47) As shown in FIG. **2**, coil springs **55** (an example of a biasing member) are disposed in front of the thermal head **28**. The coil spring **55** is in contact with the thermal head **28** at a rear end and is also in contact with the internal frame at a front end (also refer to FIG. **19**). The coil spring **55** biases the thermal head **28** to the platen roller **10** in printing, whereby the thermal head **28** is pressed against the platen roller **10** by an optimum pressure for printing.
(48) The printer **1** includes a peeling unit **4** and performs continuous issuing and peeling issuing in accordance with the peeling unit **4** moved between a continuous issuing position and a peeling issuing position. As shown in FIG. **1**B, a peeling unit open button **52***b* is exposed when the printer cover **3** is at the open position. The peeling unit **4** is moved by operating the peeling unit open button **52***b*. FIG. **2** shows a state of the peeling unit **4** when the peeling unit open button **52***b* is operated.
(49) As described later, the peeling unit open button **52***b* is operated by an operator, in order to switch from continuous issuing to peeling issuing.
(50) As shown in FIG. **2**, the peeling unit **4** includes a peeling roller cover **41** and a peeling roller holder **42** that holds a peeling roller **45**. The peeling roller cover **41** covers the peeling roller holder **42** in continuous issuing. The peeling roller cover **41** is axially supported by the internal frame in the body case **2** and swings from a closed position to an open position (state shown in FIG. **2**) in accordance with operation to the peeling unit open button **52***b*.
(51) The peeling roller holder **42** is axially supported by the peeling roller cover **41**. In continuous issuing, the peeling roller holder **42** is contained in such a manner as to be folded under a back surface of the peeling roller cover **41**.
(52) The peeling unit **4** will be detailed later.
(53) The printer cover **3** is provided with a sensor **35**. The sensor **35** is disposed in a feeding path of the continuous paper "P", along which the continuous paper "P" pulled out of the paper roll "R" reaches the platen roller **10**. The sensor **35** detects positions of labels PL, when the printer cover **3** is in the closed state. It is preferable to control a feeding amount of the continuous paper "P" based on results detected by the sensor **35**.
(54) Although not shown, it is preferable to provide a cutter for cutting the liner PM of the

continuous paper "P" that has been continuously issued. In the case of providing a cutter, the cutter is placed at the ejection part **20** so as to extend along the width direction of the continuous paper "P." Alternatively, the function of the cutter may be imparted to the peeling bar **12**.

Continuous Issuing and Peeling Issuing

(55) Next, continuous issuing and peeling issuing of the printer **1** will be described with reference to FIG. **3**.

(56) The printer **1** is configured to allow switching between peeling issuing and continuous issuing. Peeling issuing is issuing labels after peeling them from a liner of a continuous paper, while continuous issuing is issuing labels without peeling them from the liner.

(57) For continuous issuing, a liner that is attached with a necessary amount of labels is prepared, and the labels can be affixed by peeling them from the liner in a working site. Thus, continuous issuing is appropriate for a situation that a target on which a label is to be affixed is distant from the printer **1**. In order to perform continuous issuing, the peeling unit **4**, which is mounted to the printer **1**, is set to the continuous issuing position.

(58) On the other hand, in the case of peeling issuing, labels are ejected one by one in a state of being peeled from a liner. Thus, peeling issuing is appropriate for a situation that a target on which a label is to be affixed is close to an operator. In order to perform peeling issuing, the peeling unit **4**, which is mounted to the printer **1**, is set to the peeling issuing position. In this state, as a continuous paper is fed by rotating the platen roller **10** in order to perform printing, while a liner is fed in a state of being nipped between the peeling roller **45** and the platen roller **10**, printed labels are individually peeled from the liner and are then ejected to the outside of the printer **1**.

(59) FIG. **3** shows schematic partial sectional views showing positional relationships between the peeling unit **4**, the platen roller **10**, the peeling bar **12**, and the thermal head **28** in continuous issuing and in peeling issuing. The peeling roller cover **41** and the peeling roller holder **42** of the peeling unit **4** are represented only by outlines in FIG. **3**. The outline of the peeling roller cover **41** is shown by a dotted line.

(60) In addition, the position of the peeling roller holder **42** differs between continuous issuing and peeling issuing, and therefore, only the peeling roller holder **42** is shown by hatching.

(61) The position of the peeling unit **4** in continuous issuing corresponds to the continuous issuing position, whereas the position of the peeling unit **4** in peeling issuing corresponds to the peeling issuing position.

(62) As shown in FIG. **3**, in continuous issuing, the peeling roller holder **42** is contained under the peeling roller cover **41**, and the peeling roller **45** is thereby at a position spaced apart the platen roller **10** and thus does not interrupt ejection of the continuous paper "P." The continuous paper "P" that has been pulled out of the paper roll "R" is nipped between the platen roller **10** and the thermal head **28**, and labels on the continuous paper "P" are printed.

(63) In order to switch from continuous issuing to peeling issuing, the peeling roller holder **42** is swung around a shaft **42***a* to a position shown in FIG. **3**. As shown in FIG. **3**, in peeling issuing, the peeling roller **45** is disposed at a position facing the platen roller **10**. Also, in peeling issuing, the continuous paper "P" that has been pulled out of the paper roll "R" is nipped between the platen roller **10** and the thermal head **28**, and labels on the continuous paper "P" are printed. This movement is the same as in continuous issuing. In peeling issuing, the liner PM of the continuous paper "P" that has been pulled out of the paper roll "R" is quickly turned by the peeling bar **12** and is nipped between the platen roller **10** and the peeling roller to be ejected. In accordance with quick turning of the liner PM at the peeling bar **12**, a label PL is peeled from the liner PM and ejected.

Opening Movement of Printer Cover **3**

(64) Next, opening movement of the printer cover **3** will be described with reference to FIGS. **4** and **5**. In addition, a cover open lever **51** and a peeling unit open lever **52** will also be described.

(65) FIG. **4** shows side views of the cover open lever **51**, the peeling unit open lever **52**, the platen-holding bracket **27**, and the peeling unit **4** when the printer cover is closed and when the cover open

button is operated. FIG. **4** shows an exemplary situation in which the peeling unit **4** is at the continuous issuing position.
(66) As shown in FIG. **4**, in a side view, the cover open lever **51** and the peeling unit open lever **52** are disposed to face in the front-rear direction, while extending in the front-rear direction at mutually different heights, resulting in space-efficient arrangement.
(67) FIG. **5** is a perspective rear view of the cover open lever **51**, the peeling unit open lever **52**, the platen-holding bracket **27**, and the peeling unit **4** when the printer cover is closed. FIG. **5** omits illustration of the peeling unit **4**.
(68) The cover open lever **51** has the cover open button **51***b* that is exposed to the outside, as shown in FIG. **1**A. The cover open lever **51** is formed with a shaft insertion hole **51***a*, and a shaft part **56** (not shown in FIG. **5**) that is provided to the internal frame is inserted in the shaft insertion hole **51***a*. This makes the cover open lever **51** be able to swing around the shaft part **56**. As shown in FIG. **5**, the cover open lever **51** has a protrusion **51***c* that protrudes inward.
(69) As shown in FIG. **5**, the platen-holding bracket **27** has a shaft **27***a*. One end of the shaft **27***a* is inserted in a boss **52***a* that is provided to the peeling unit open lever **52**, whereas the other end of the shaft **27***a* is inserted in a boss that is provided to the internal frame (not shown). This makes the platen-holding bracket **27** be able to swing around the shaft **27***a*.
(70) In addition, the peeling unit open lever **52** has an engaging protrusion **523** (refer to FIG. **4**) that protrudes inward, although the engaging protrusion **523** is not seen in FIG. **5**. As described later, the engaging protrusion **523** engages with the peeling roller cover **41** of the peeling unit **4**.
(71) The platen-holding bracket **27** has a hole **27***c* that is formed in a side wall, and the protrusion **51***c* of the cover open lever **51** is inserted in the hole **27***c*. Herein, the hole **27***c* is formed greater than the protrusion **51***c* in a side view (that is, the hole **27***c* has play), whereby the platen-holding bracket **27** is able to swing. The platen-holding bracket **27** swings around the shaft **27***a*, whereas the cover open lever **51** swings around the shaft part **56** (refer to FIG. **4**), and therefore, they have different swing axes. In consideration of this, the hole **27***c* is provided with play so as to absorb the difference in trajectory between the hole **27***c* and the protrusion **51***c* due to the different swing axes.
(72) The peeling unit open lever **52** is able to turn (or swing) around the shaft **27***a*, which is inserted in the boss **52***a*. That is, the platen-holding bracket **27** and the peeling unit open lever **52** share the single swing shaft **27***a*, which eliminates a need to provide another swing shaft for the peeling unit open lever **52**, resulting in contribution to reduction in space and cost. Nevertheless, the structure is not limited thereto, and in another embodiment, an individual swing shaft may be set to each of the platen-holding bracket **27** and the peeling unit open lever **52**.
(73) A coil spring **53** is interposed between the peeling unit open lever **52** and the internal frame (not shown), at a position immediately below the peeling unit open button **52***b*. Upon being pressed (operated) down against a restoring force of the coil spring **53**, the peeling unit open lever **52** swings around the shaft **27***a* (swings in a clockwise direction in FIG. **4**). As described later, in accordance with swinging of the peeling unit open lever **52**, the peeling unit **4** swings via the engaging protrusion **523** and moves from the closed position to the open position.
(74) When the force for pressing down the peeling unit open button **52***b* is released, the peeling unit open lever **52** returns (swings) to the position where it is disposed before being pressed down, by the restoring force of the coil spring **53**.
(75) The platen-holding bracket **27** is biased by a pair of coil springs **29**. In FIG. **5**, one end of each of the coil springs **29** is hooked to the platen-holding bracket **27**, whereas the other end of each of the coil springs **29** is hooked to the internal frame (not shown).
(76) Unless an external force is applied to the platen-holding bracket **27**, the platen-holding bracket **27** is in the position at the time the printer cover is closed, as shown in FIG. **4**, and it holds the platen shaft **10***a* in a groove **27***b*. This position is a locking position for locking the printer cover **3** coupled to the platen shaft **10***a*, at the closed position.
(77) In this state, in response to the cover open button **51***b* being pressed (operated) down, the

cover open lever **51** swings around the shaft part **56** (swings in a counterclockwise direction in FIG. **4**). In accordance with swinging of the cover open lever **51**, the protrusion **51***c* presses a rim of the hole **27***c* of the platen-holding bracket **27** to swing the platen-holding bracket **27** around the shaft **27***a* (in the clockwise direction in FIG. **4**) against the restoring force of the coil spring **29**.

(78) As described above, the printer cover **3**, which is mounted with the platen shaft **10***a*, is biased in the direction from the closed position to the open position. Thus, the printer cover **3** moves to the open position when the platen shaft **10***a* comes off from the groove **27***b* due to swinging of the platen-holding bracket **27**. The position of the platen-holding bracket **27** at this time is an unlocking position for unlocking the printer cover **3** at the closed position.

(79) Conversely, in closing the printer cover **3**, a pressing down force of an operator closing the printer cover **3** makes the platen shaft **10***a*, which is mounted to the printer cover **3**, press down an inclined top part of the platen-holding bracket **27** against the restoring force of the coil spring **29**. In response to this, the platen-holding bracket **27** is swung in the clockwise direction in FIG. **4**, and the platen shaft **10***a* is inserted in the groove **27***b* of the platen-holding bracket **27**. In the state in which the platen shaft **10***a* is inserted in the groove **27***b*, the platen-holding bracket **27** returns to the locking position at the time the printer cover is closed, as shown in FIG. **4**, by the restoring force of the coil spring **29**.

Peeling Unit **4**

(80) Next, the peeling unit **4** will be described with reference to FIGS. **6** to **9**.

(81) FIG. **6** shows perspective views of the peeling unit **4** when open and when closed. The peeling unit **4** when closed, which is shown in FIG. **6**, is at the continuous issuing position.

(82) The open position of the peeling unit **4** is a position when the peeling roller cover **41** is opened in accordance with operation to the peeling unit open button **52***b*. That is, the open position of the peeling unit **4** corresponds to the open position of the peeling roller cover **41**.

(83) The open position of the peeling roller cover **41** is a position for exposing at least a part of the inside of the printer **1** or at least a part of the inside of the body case **2**, as shown in FIG. **2**. In one example, as shown in FIG. **2**, when the peeling roller cover **41** is at the open position, the coil springs **55**, a flexible cable **57** (refer to FIG. **21**) connected to the thermal head **28**, and so on, inside the body case **2**, are exposed. From another point of view, the open position of the peeling roller cover **41** may be defined as a position for exposing the thermal head **28** inside the body case **2**. From yet another point of view, the open position of the peeling roller cover **41** may be also defined as a position for allowing opening the peeling roller cover **41** at the corresponding position when the peeling roller holder **42** is at a position facing the back surface of the peeling roller cover **41**. As shown in FIG. **6**, when the peeling roller cover **41** is at the open position (is open), the peeling roller holder **42** protrudes upward (in a protruding state).

(84) The closed position of the peeling unit **4** is a position when the peeling roller cover **41** is closed. That is, the closed position of the peeling unit **4** corresponds to the closed position of the peeling roller cover **41**.

(85) The closed position of the peeling roller cover **41** is a position for covering at least a part of the inside of the printer **1** or at least a part of the inside of the body case **2**, which is exposed when the peeling roller cover **41** is at the open position. In one example, as shown in FIGS. **1**A and **1**B, when the peeling roller cover **41** is at the closed position, the coil springs **55**, the flexible cable **57**, and so on, are not visible from the outside and are covered. From another point of view, the closed position of the peeling roller cover **41** may be defined as a position for covering at least a part of the thermal head **28** inside the body case **2**. From yet another point of view, the closed position of the peeling roller cover **41** may be also defined as a position for retaining the peeling roller cover **41** at the corresponding position instead of opening it, when the peeling roller holder **42** is at the position facing the back surface of the peeling roller cover **41**. When the peeling roller cover **41** is at the closed position, the position of the peeling roller holder **42** differs between for continuous issuing and for peeling issuing.

(86) As shown in FIG. **6**, when the peeling roller cover **41** is at the closed position in continuous issuing, the peeling roller holder **42** is contained under the peeling roller cover **41** (in a contained state).
(87) With reference to FIG. **6**, the peeling roller cover **41** is a swing member that has a pair of shafts **41***a* and is thereby able to swing around the shafts **41***a*. The shaft **41***a* has a circular cross section and is inserted in a tubular part (not shown), which is provided to the internal frame, so as to be rotatable. The tubular part is preferably formed with, for example, an elongated hole in the printer front-rear direction so that the shaft **41***a* can be slightly displaced in the printer front-rear direction thereinside. The elongated hole provides play in the printer front-rear direction to the shaft **41***a* that is inserted in the tubular part, and it improves resistance to impact of falling, etc., of the printer **1**.
(88) The direction of the elongated hole that is formed in the tubular part is not limited to the printer front-rear direction, and for example, it can be set to any direction such as the upper-lower direction of the printer **1**, in a plane perpendicular to the right-left direction of the printer **1**.
(89) The peeling roller cover **41** extends in the same direction as the extending direction of the platen roller **10**. The peeling roller cover **41** has a surface **411** and a back surface **412**. The surface **411** is a surface that is exposed when the peeling roller cover **41** is at the closed position. The back surface **412** is formed with a recess so as to contain the peeling roller holder **42**. Conversely, the surface **411** has a swollen shape at the center in the front-rear direction, which is convenient to cut a liner when a cutter is provided to the ejection part **20**.
(90) The peeling roller cover **41** is formed with an engaging hole **415** in the vicinity of the shaft **41***a*. As described later, the engaging protrusion **523** of the peeling unit open lever **52** is inserted in the engaging hole **415**.
(91) The peeling roller cover **41** may be provided at the side with a pair of U-shaped grooves **413**. The U-shaped groove **413** abuts on a protrusion **26** (refer to FIG. **18**) that is formed to the internal frame, when the shaft **41***a* is at the closed position. The U-shaped groove **413** functions as a part for positioning in the upper-lower direction of the peeling unit **4** by abutting on the protrusion **26**. The U-shaped groove **413** in the state of abutting on the protrusion **26** provides a predetermined gap between the peeling unit **4** and the thermal head **28**. Thus, it is possible to reliably prevent interference between the peeling unit **4** and the thermal head **28**.
(92) As described above, the shaft **41***a* of the peeling unit **4** is preferably inserted in the elongated hole in the printer front-rear direction, which is formed in the tubular part of the internal frame, whereby play is provided in the printer front-rear direction. Under these conditions, the U-shaped groove **413** in the state of abutting and being engaged with the protrusion **26** (refer to FIG. **18**) prevents positional deviation in the printer front-rear direction of the peeling unit **4** (peeling roller cover **41**) due to play of the shaft **41***a* inserted in the elongated hole (that is, functions as a part for positioning in the printer front-rear direction of the peeling unit **4**).
(93) However, the U-shaped groove **413** and the protrusion **26** are not necessarily provided. The abutting parts of a part of the peeling unit **4** and the internal frame can be formed into any shape as appropriate, in the condition in which they can abut on each other while ensuring the gap between the peeling unit **4** and the thermal head **28**. Instead of such an abutting structure, the position in the upper-lower direction of the peeling unit **4** can be determined by, for example, limiting the movable range of the shaft **41***a* of the peeling roller cover **41**.
(94) The surface **411** of the peeling roller cover **41** is disposed with a peeling sensor **47**. The peeling sensor **47** is an optical reflective sensor that detects presence or absence of a label peeled in peeling issuing. With reference to FIG. **3**, a label PL that is peeled by the peeling bar **12** is controlled so that its part on a feeding direction upstream side will be fed and stop in the vicinity of the peeling bar **12**, and the peeled label PL thereby remains at the peeling bar **12** by its adhesive strength. The peeling sensor **47** detects presence or absence of this label PL. When the peeled label is picked up by an operator, the peeling sensor **47** detects absence of the label PL, and control is

performed to issue a next label.
(95) With reference to FIG. **6**, the peeling roller holder **42** is a member that holds the peeling roller **45**.
(96) The peeling roller holder **42** extends in the same direction as the extending direction of the platen roller **10**, as in the case of the peeling roller cover **41**. The peeling roller holder **42** is configured to be contained under the back surface **412** of the peeling roller cover **41**. For this purpose, a pair of shafts **42***a* are disposed inward of the pair of shafts **41***a* of the peeling roller cover **41**, and the width of the peeling roller holder **42** is made smaller than that of the peeling roller cover **41**.
(97) The peeling roller holder **42** is a swing member that has the pair of shafts **42***a* and is thereby able to swing around the shafts **42***a*. The pair of shafts **42***a* are axially supported by the peeling roller cover **41**, at positions separated from the shafts **41***a*. The peeling roller **45** is disposed to distal ends of arms **421** extending from the shafts **42***a*. Thus, as shown in FIG. **6**, the peeling roller **45** largely protrudes upward based on the shafts **41***a*, when the peeling unit **4** is open.
(98) That is, the peeling roller holder **42** is able to swing between a contained position when the peeling roller **45** is contained under the peeling roller cover **41**, and a protruding position when the peeling roller **45** is not covered with the peeling roller cover **41**. The protruding position is a position when the peeling roller cover **41** is open, as shown in FIG. **6**. The contained position of the peeling roller holder **42** is also a position for facing the back surface **412** of the peeling roller cover **41** as well as a position for being covered with the peeling roller cover **41**.
(99) In order to contain the peeling roller holder **42**, the peeling roller holder **42** is swung around the shaft **42***a* to the back surface **412** of the peeling roller cover **41**, and moreover, the whole peeling roller cover **41** and peeling roller holder **42** are swung around the shaft **41***a*. As a result, the peeling roller holder **42** is compactly contained under the peeling roller cover **41** in such a manner as to be folded down.
(100) On the other hand, when the peeling roller cover **41** is at the open position, the peeling roller holder **42** is able to swing between the contained position and the protruding position. As described later, the peeling roller holder **42** is biased in a direction from the contained position to the protruding position by a coil spring **43**. Thus, immediately after the peeling roller cover **41** moves from the closed position to the open position, the peeling roller holder **42** moves in such a manner as to spring out from the contained position to the protruding position. This structure enables an operator to quickly switch from continuous issuing to peeling issuing.
(101) When the peeling roller holder **42** is at the protruding position, the peeling roller **45** is highly protruded. This enables moving the peeling roller **45** to a distant position in setting the peeling unit **4** to the peeling issuing position.
(102) A pair of arms **421** extend from the pair of shafts **42***a*. A shaft **45***a* for rotating the peeling roller **45** and auxiliary rollers **46** is disposed at the distal ends of the pair of arms **421**. Each of the auxiliary rollers **46** has a diameter smaller than that of the peeling roller **45**. Providing the auxiliary rollers **46** on both sides of the peeling roller **45** enables smoothly ejecting a wide liner in peeling issuing of a wide label. If the auxiliary rollers **46** were not provided, a wide liner would be able to move in the width direction (right-left direction); but providing the auxiliary rollers **46** enables stably feeding a wide liner.
(103) However, the auxiliary rollers **46** are not necessarily provided. In the case of not using the auxiliary rollers **46**, peeling issuing can be executed in the condition in which the peeling roller **45** is provided.
(104) Each of the arms **421** is formed with a protrusion **422** that protrudes outward. As described later, the protrusion **422** is provided so as to engage the peeling unit **4** with the printer cover **3** in peeling issuing.
(105) As shown in FIG. **6**, a pair of coil springs **43** are provided in the vicinity of the pair of shafts **42***a* of the peeling roller holder **42**. Although not shown, the coil spring **43** is coupled to the peeling

roller holder **42** at one end and is also coupled to the peeling roller cover **41** at the other end, and it thereby biases the peeling roller holder **42** in the direction for swinging from the contained position to the protruding position. With this structure, when the peeling roller cover **41** is at the open position (that is, the peeling unit **4** is at the open position), the peeling roller holder **42** is at the protruding position at any time.

(106) FIG. **7** is a perspective view of the peeling unit **4** when open, as seen from a viewpoint different from that of FIG. **6**. In the state in which the peeling roller holder **42** is at the protruding position, the arms **421** of the peeling roller holder **42** partially abut on the surface **411** of the peeling roller cover **41**. In other words, the surface **411** of the peeling roller cover **41** functions as a stopper for the peeling roller holder **42** that is swung by the coil spring **43**.

(107) Next, movements in making the peeling unit **4** be at the open position from the state in continuous issuing, will be described with reference to FIGS. **8** and **9**.

(108) FIGS. **8** and **9** sequentially show side views of the peeling unit open lever **52** and the peeling unit **4**, from states S**1** to S**3**.

(109) The state S**1** shows a state in which the printer cover **3** is open in continuous issuing. The state S**2** shows a state of continuously operating the peeling unit open button. The state S**3** shows a state of releasing operation of the peeling unit open button.

(110) The peeling unit open lever **52** and the peeling unit **4** are engaged with each other by inserting the engaging protrusion **523** of the peeling unit open lever **52** in an engaging hole **415** of the peeling roller cover **41**, from the inside. The peeling unit open lever **52** swings so as to move the peeling roller cover **41** between the closed position and the open position.

(111) The engaging hole **415** has, for example, a heart shape, and it allows the engaging protrusion **523** to move therein.

(112) As shown by the state S**1** in FIG. **8**, when the printer cover **3** is open in continuous issuing, the engaging protrusion **523** is positioned on a lower side in the engaging hole **415**. In this state, the peeling roller holder **42** is at the contained position under the back surface **412** (refer to FIG. **6**) of the peeling roller cover **41**.

(113) When the peeling unit open button **52***b* is pressed (operated) down, the peeling unit open lever **52** swings around the shaft **27***a* in a clockwise direction in FIG. **8**. In response to this, the engaging protrusion **523** of the peeling unit open lever **52** moves upward in the engaging hole **415** and upwardly presses the peeling roller cover **41**, at an upper rim of the engaging hole **415**. The peeling roller cover **41** is thereby swung around the shaft **41***a* to the open position in a counterclockwise direction in FIG. **8**. As described above, the peeling roller holder **42** is biased in the direction for swinging from the contained position to the protruding position, by the coil spring **43** (refer to FIG. **6**). Thus, as the peeling roller cover **41** swings to the open position, a position restriction of the peeling roller holder **42** due to a second stopper **522** is released to form space in which the peeling roller holder **42** is able to swing. As a result, the peeling roller holder **42** swings to the protruding position, as shown by the state S**2** in FIG. **8**. The second stopper **522** will be described later.

(114) As shown by the state S**2** in FIG. **8**, the position of the shaft **42***a* is higher when the peeling roller cover **41** is at the open position than when the peeling roller cover **41** is at the closed position. In addition, as described above, in the printer **1**, space is formed in which the peeling roller holder **42** is able to swing from the contained position to the protruding position, when the peeling roller cover **41** is at the open position. Thus, the peeling roller holder **42** springs up by the biasing force of the coil spring **43**.

(115) When pressing down of the peeling unit open button **52***b* is released from the state shown by the state S**2**, the peeling unit open lever **52** swings around the shaft **27***a* in the counterclockwise direction in FIG. **8**, with the restoring force of the coil spring **53**. The peeling unit open lever **52** and the peeling roller cover **41** thereby return to the positions in the state S**1**. Meanwhile, the peeling roller holder **42**, which swings to the protruding position once, remains at the protruding

position, instead of returning to the contained position. As a result, the peeling unit **4** is in the condition shown by the state S**3** in FIG. **9**.

Engagement Between Peeling Unit **4** and Printer Cover **3**

(116) In the printer **1**, the peeling unit **4** is set to the peeling issuing position while the printer cover **3** and the peeling unit **4** are engaged with each other, by swinging the printer cover **3** from the open position to the closed position in the state S**3** in FIG. **9**.

(117) Hereinafter, engagement between the peeling unit **4** and the printer cover **3** in peeling issuing will be described with reference to FIGS. **10**A, **10**B, and **11**.

(118) First, the structure of the printer cover **3** for engaging with the peeling unit **4** will be described with reference to FIGS. **10**A and **10**B. FIG. **10**A is a plane view of the printer cover **3**, and FIG. **10**B is an enlarged view of an A-A cross section in FIG. **10**A.

(119) As shown in FIG. **10**A, the printer cover **3** has a pair of peeling unit-receiving parts **31** at front ends. The peeling unit-receiving part **31** is provided in the vicinity of the position at which the platen roller **10** and the peeling bar **12** are supported.

(120) As shown in FIG. **10**B, the peeling unit-receiving part **31** is formed with a guide groove **31***p* that opens forward. The guide groove **31***p* is a groove that opens only to the inside along a direction from the front end to a rear end of the printer cover **3**. The guide groove **31***p* receives the protrusion **422** (refer to FIG. **9**) of the peeling unit **4** that is positioned on a front side, in the process of closing the printer cover **3**.

(121) A roller-pressing mechanism **37** is provided in the guide groove **31***p*. As described later, the roller-pressing mechanism **37** presses the peeling roller **45** to the platen roller **10** to generate a nip pressure for nipping a liner between the peeling roller **45** and the platen roller **10**, when the printer cover **3** is at the closed position.

(122) The roller-pressing mechanism **37** includes an abutting part **32** that is disposed in the guide groove **31***p* and also includes a coil spring **33** that is disposed behind the abutting part **32**. In accordance with the printer cover **3** being moved to the closed position, the protrusion **422** of the peeling unit **4** is guided to the abutting part **32**.

(123) When operation to move the printer cover **3** from the open position to the closed position is performed in the state S**3** in FIG. **9**, the protrusion **422** of the peeling unit **4** enters the guide groove **31***p* of the peeling unit-receiving part **31** during the process of moving the printer cover **3**. As the printer cover **3** swings to the closed position, the protrusion **422** advances toward the rear of the printer cover **3** along the guide groove **31***p* and abuts on the abutting part **32**. In this manner, the printer cover **3** engages with the peeling unit **4**. When the printer cover **3** reaches the closed position, the peeling roller **45** of the peeling unit **4** engaging with the printer cover **3** is at a position facing the platen roller **10**.

(124) Thus, it is possible for an operator to engage the printer cover **3** with the peeling unit **4** while moving the peeling unit **4** to the peeling issuing position, only by operation to close the printer cover **3**.

(125) FIG. **11** is an enlarged sectional view showing a part in the vicinity of the platen roller **10** when the printer cover **3** is completely closed and the peeling unit **4** is set to the peeling issuing position.

(126) As shown in FIG. **11**, when the printer cover **3** is at the closed position, the peeling roller **45** of the peeling unit **4** is disposed at a position facing the platen roller **10**. In this state, the protrusion **422** of the peeling unit **4** abuts on the abutting part **32** of the peeling unit-receiving part **31** of the printer cover **3** to compress the coil spring **33** behind the abutting part **32**. A restoring force of the coil spring **33** acts on the peeling roller **45** via the protrusion **422** and thereby makes the peeling roller **45** press the platen roller **10**, resulting in generation of a nip pressure for nipping a liner. With this structure, a force in a rotation direction around the shaft **42***a* of the peeling roller holder **42** (F**5***c* in FIG. **11**) is converted into a nip pressure between the peeling roller **45** and the platen roller **10**.

(127) In an embodiment, a normal line direction of an abutting surface of the abutting part **32** abutted with the protrusion **422** (direction denoted by a reference symbol "F**5***b*"), and a direction from the center of the peeling roller **45** to the center of the platen roller **10**, may be the same in a side view, as shown in FIG. **11**. However, the direction of the force F**5** varies depending on the abutting angle between the protrusion **422** and the abutting part **32**, and therefore, these directions may not be the same. As shown in FIG. **11**, a component force F**5***b* being a normal component with respect to the abutting surface, of a reaction force F**5** of the abutting part **32** acting on the protrusion **422**, causes the peeling roller **45** to press the platen roller **10**, whereby a nip pressure for nipping a liner is more effectively generated.

Movement for Containing Peeling Roller Holder **42**

(128) Next, movement for moving the peeling roller holder **42** at the protruding position to contain it under the peeling roller cover **41** and setting the peeling unit **4** to the continuous issuing position, will be described with reference to FIGS. **12** and **13**.

(129) In order to switch from peeling issuing to continuous issuing, the cover open button **51***b* is pressed down to open the printer cover **3**, and the peeling unit open button **52***b* is then pressed down. In response to this, as shown by the state S**2** in FIG. **8**, the peeling roller cover **41** swings to the open position, and the peeling roller holder **42** swings to the protruding position. In this state, operation to fold down the peeling roller holder **42** to contain it under the peeling roller cover **41** (folding operation) is performed by an operator, whereby the peeling unit **4** is set to the continuous issuing position.

(130) FIGS. **12** and **13** sequentially show perspective views of the peeling unit open lever **52** and the peeling unit **4** when an operator performs the folding operation of the peeling unit **4**, from states S**5** to S**9**.

(131) As shown in FIG. **12**, the peeling unit open lever **52** has a first stopper **521** and a second stopper **522** that protrude inward. The first stopper **521** and the second stopper **522** are disposed separately in the front-rear direction and are provided so as to abut on the arm **421** of the peeling roller holder **42** and thereby restrict swinging of the arm **421**.

(132) The state S**5** in FIG. **12** is a state in which the peeling roller cover **41** is at the open position and the peeling roller holder **42** is at the protruding position, which corresponds to the state S**2** in FIG. **8**. An operator can maintain this state by continuously pressing down the peeling unit open button **52***b*.

(133) In the state S**5**, an operator may rotate (or swing) the peeling roller holder **42** around the shaft **42***a* and move it to the contained position under the back surface **412** of the peeling roller cover **41**. Thus, the state is changed to the state S**6**. At this time, a part most distant from the shaft **42***a* of the arm **421** crosses over the first stopper **521** by the operating force of the operator. This makes the arm **421** abut on the first stopper **521** to restrict swinging of the peeling roller holder **42**, against the restoring force of the coil spring **43** (refer to FIG. **6**). That is, when the peeling roller holder **42** is in the contained position and the peeling roller cover **41** is at the open position, the first stopper **521** abuts on the arm **421** to restrict swinging of the peeling roller holder **42**.

(134) In the state in which the first stopper **521** restricts swinging of the peeling roller holder **42**, it is easy to move the peeling roller cover **41** to the closed position while retaining the peeling roller holder **42** at the contained position. If the first stopper **521** were not provided, an operator would need to move the peeling roller cover **41** to the closed position by releasing pressing down the peeling unit open button **52***b* while holding the peeling roller holder **42** by hand so as to prevent it from swinging from the contained position. Thus, providing the first stopper **521** improves operability.

(135) When the operator releases pressing down the peeling unit open button **52***b* in the state in which the peeling roller holder **42** is locked at the contained position by the first stopper **521**, the peeling roller cover **41** starts moving to the closed position. The state S**7** shows a state while the peeling roller cover **41** is moving to the closed position.

(136) In the process in which the peeling roller cover **41** moves to the closed position, restriction of swinging of the arm **421** by the first stopper **521** is released in accordance with swinging of the peeling roller cover **41**. Specifically, an outer edge of the arm **421** is formed so that restriction of swinging of the arm **421** will be released at the time the peeling roller cover **41** is closed to a predetermined angle.
(137) The state S**8** in FIG. **13** is a state at the time the operator further closes the peeling roller cover **41** from the state S**7**. The peeling roller holder **42**, in which restriction of swinging by the first stopper **521** is released, is swung by the restoring force of the coil spring **43**, but it is again restricted from swinging by the second stopper **522**, which is on a rear side of the first stopper **521**. That is, the second stopper **522** comes into contact with the arm **421** while the peeling roller holder **42** moves from the open position to the closed position, whereby it restricts the peeling roller holder **42** from swinging between the contained position and the protruding position. The state S**9** is a state in which the peeling roller cover **41** is at the closed position and the peeling unit **4** is at the continuous issuing position.
(138) Providing the second stopper **522** prevents the peeling roller holder **42** from swinging while the peeling roller cover **41** moves from the open position to the closed position. Moreover, the second stopper **522** is positioned rearward of the first stopper **521**, and thus, when the peeling unit open button **52***b* is operated in the state in which the peeling unit **4** is at the continuous issuing position as shown by the state S**9**, the peeling roller holder **42** smoothly swings to the protruding position.
(139) The first stopper **521** and the second stopper **522** are not necessarily provided. Providing even only one of the stoppers can contribute to improving operability. It is also possible to perform the folding operation of the peeling unit **4**, even when both of the first stopper **521** and the second stopper **522** are not provided. Specifically, it is possible for an operator to contain the peeling roller holder **42** under the peeling roller cover **41** by carefully moving the peeling roller cover **41** to the closed position while holding the peeling roller holder **42** at the contained position by hand.
Movement to Switch Between Continuous Issuing and Peeling Issuing of Printer **1**
(140) Next, movement to switch between continuous issuing and peeling issuing of the printer **1** will be described with reference to FIGS. **14** and **15**.
(141) FIGS. **14** and **15** sequentially show side views of a main part of the printer **1** at the time of switching from continuous issuing to peeling issuing, from states S**10** to S**15**. FIG. **15** omits illustration of the platen-holding bracket **27**.
(142) The state S**10** in FIG. **14** shows a state of the printer **1** in continuous issuing. In this state, the platen shaft **10***a* of the platen roller **10**, which is axially supported by the printer cover **3**, is fitted in the groove **27***b* of the platen-holding bracket **27**, whereby the printer cover **3** is held. In the state S**10**, the peeling unit **4** is set to the continuous issuing position.
(143) When an operator presses down the cover open button **51***b* in the state S**10**, holding of the platen shaft **10***a* by the platen-holding bracket **27** is released. Then, as shown by the state S**11**, the printer cover **3** is moved to the open position by the biasing force of the torsion spring provided to the hinge **8** (refer to FIG. **2**).
(144) Subsequently, when the operator presses down the peeling unit open button **52***b*, the peeling roller cover **41** swings from the closed position to the open position, and the peeling roller holder **42** swings from the contained position to the protruding position, as shown by the state S**12**. Then, when the operator releases pressing down of the peeling unit open button **52***b*, the peeling roller cover **41** returns to the closed position, but the peeling roller holder **42** remains at the protruding position with the biasing force of the coil spring **43** (refer to FIG. **6**), as shown by the state S**13** in FIG. **15**.
(145) Next, in accordance with the printer cover **3** being closed by the operator, the protrusion **422** of the peeling roller holder **42** at the protruding position is inserted in the guide groove **31***p* (refer to FIG. **10**B) of the printer cover **3** and is guided therealong, whereby the printer cover **3** and the

peeling unit **4** engage with each other, as shown by the state S**14**.

(146) As shown by the state S**15**, when the printer cover **3** reaches the closed position, the platen shaft **10***a* of the platen roller **10** is held by the platen-holding bracket **27**, and the peeling unit **4** is set to the peeling issuing position. That is, the peeling roller **45** of the peeling unit **4** is disposed at the position facing the platen roller **10** to nip the liner PM with the platen roller **10**. In this state, as described above, the protrusion **422** that is engaged with the printer cover **3** is pressed by the coil spring **33** (refer to FIG. **11**), whereby an appropriate nip pressure against the platen roller **10** is generated in the peeling roller **45**.

(147) In peeling issuing, a label PL that is printed by the thermal head **28** is peeled from the liner PM, due to the liner PM being quickly turned by the peeling bar **12**. The peeling roller **45** is driven to rotate in accordance with rotation of the platen roller **10** and ejects the liner PM.

(148) In order to switch from peeling issuing to continuous issuing, the cover open button **51***b* is pressed down to open the printer cover **3**, and the peeling unit open button **52***b* is then pressed down. This causes the peeling roller cover **41** of the peeling unit **4** to swing to the open position and also causes the peeling roller holder **42** to swing to the protruding position. Thereafter, as described with reference to FIGS. **12** and **13**, the folding operation of the peeling unit **4** is performed to set the peeling unit **4** to the continuous issuing position.

(149) As described above, the printer **1** of the embodiment includes the peeling unit **4** that is movable between the continuous issuing position and the peeling issuing position. When the peeling unit **4** is at the continuous issuing position, the peeling roller holder **42** holding the peeling roller **45** is compactly contained at the contained position under the back surface of the peeling roller cover **41**.

(150) Switching from continuous issuing to peeling issuing is performed by a simple operation as follows: opening the printer cover **3**; operating the peeling unit open button **52***b* to move the peeling roller holder **42** to the protruding position; and closing the printer cover **3**. That is, switching can be performed by a simple action of these easy three steps, and operability is excellent. In addition, when the printer cover **3** is at the closed position, the peeling roller **45** is pressed against the platen roller **10** by the roller-pressing mechanism **37** of the printer cover **3**, resulting in generation of an appropriate nip pressure.

(151) Conversely, switching from peeling issuing to continuous issuing is performed as follows: opening the printer cover **3**; operating the peeling unit open button **52***b* to move the peeling roller holder **42** to the protruding position; performing the folding operation to move the peeling roller holder **42** to the contained position; and closing the printer cover **3**. Also in this case, the operation is simple.

Method of Mounting and Removing Thermal Head **28**

(152) Next, a method of mounting and removing the thermal head **28** to and from the printer **1** will be described with reference to FIGS. **16**A to **19**.

(153) FIG. **16**A shows a front side (an example of a first side) biased by the coil spring **55**, which is one of both surfaces of the thermal head **28**, and FIG. **16**B shows a rear side (an example of a second side) of the thermal head **28**. The rear side of the thermal head **28** faces the platen roller **10**. FIG. **17** shows enlarged sectional views of an A-A cross section and a B-B cross section in FIG. **16**A.

(154) As shown in FIGS. **16**A and **17**, the thermal head **28** has a structure in which a board **282** is attached to a heat dissipation plate **281** (an example of a base) that has a substantially rectangular shape in a plane view. The heat dissipation plate **281** is made of a metal material having a high thermal conductivity, such as aluminum. The A-A cross section in FIG. **17** shows that the board **282** (an example of a coating layer) is attached to the heat dissipation plate **281** in such a manner as to extend from a surface **281***a* of the heat dissipation plate **281** to a back surface **281***b* on a side opposite to the surface **281***a*, via a first end part **281***e***1** interposed therebetween. The board **282** is, for example, a ceramic board.

(155) A region of the heat dissipation plate **281** that is occupied by the board **282** is an example of a coated region.
(156) As shown in FIG. **16**B and by the B-B cross section in FIG. **17**, a cutout **283***c* (an example of an uncoated region) is provided at a substantially center position in a longitudinal direction (lateral direction) of the surface **281***a* of the thermal head **28**. The cutout **283***c* does not have the board **282** and exposes the surface **281***a* of the heat dissipation plate **281**. As described later, the cutout **283***c* is configured to be in contact with a protrusion **211** (refer to FIG. **20**) for allowing the thermal head **28** to swing.
(157) As shown in FIG. **16**A and by the A-A cross section in FIG. **17**, the board **282** that is attached to the back surface **281***b* of the heat dissipation plate **281** is mounted with, but not limited to, surface-mount devices (SMDs) such as a connector **285**, an EEPROM **286**, and a diode **287**. In the state in which the thermal head **28** is mounted to the printer **1**, the flexible cable **57** is connected to the connector **285**. The flexible cable **57** transmits a signal from the circuit board (not shown) of the printer **1** to the thermal head **28**.
(158) In the state in which the thermal head **28** is mounted to the printer **1**, the relatively tall surface-mount devices (e.g., the connector **285**, the EEPROM **286**, and the diode **287** in FIG. **16**A), which are mounted on the back surface **281***b* of the heat dissipation plate **281**, face the front side of the printer **1**. This configuration protects these surface-mount devices from water, etc., which may enter from the ejection part **20** on a rear side of the thermal head **28**. A driver IC (not shown) is mounted in the vicinity of a heat generating part **284** on the rear side of the thermal head **28** facing the ejection part **20** (on a side on which the surface **281***a* of the heat dissipation plate **281** is provided). Due to the driver IC with low height, the driver IC and wiring are protected together with the heat generating part **284** by a protective layer or a coating layer, whereby they are unlikely to be damaged by water entering from the ejection part **20**.
(159) As shown in FIG. **16**B, the tall surface-mount devices, such as the connector, are not disposed on the rear side of the thermal head **28** (on the side disposed with the heat generating part **284**). Thus, a feed angle of a label PL relative to the heat generating part **284** can be small (in other words, it can be an angle approximately perpendicular to the heat generating part **284** in a side view) (refer to FIG. **3**). Here, good print quality is obtained due to the following reasons.
(160) The heat generating part **284** includes a glaze layer (partial graze) generally having a protrusion shape, and it thereby has a protrusion shape as a whole. If tall surface-mount devices are disposed on the rear side of the thermal head **28**, the feeding angle of a label PL relative to the heat generating part **284** is made large in order to avoid the tall surface-mount devices. In this case, due to the heat generating part **284** having a protrusion shape and to stiffness (resilience) of a label PL, the label PL tends to rise from the heat generating part **284** at the position thereof, and it is difficult to apply an appropriate printing pressure to the label PL between the heat generating part **284** and the platen roller **10**. In contrast, for a small feeding angle of a label PL relative to the heat generating part **284**, although having a protrusion shape, the heat generating part **284** pinches a label PL with the platen roller **10** by applying an appropriate printing pressure, in the vicinity of a top of the heat generating part **284**. Thus, good print quality is obtained.
(161) A pair of shafts **28***a* that extend outward are coupled to both end surfaces of the heat dissipation plate **281**. As described later, the pair of shafts **28***a* are provided in order to mount the thermal head **28** to the internal frame of the printer **1**. As shown in FIGS. **16**A and **16**B, the shaft **28***a* has a large-diameter part joined to the heat dissipation plate **281** and has a small-diameter part extending outward from the large-diameter part, and it thereby has a high strength. The small-diameter part of the shaft **28***a* is inserted in a shaft-receiving groove **25**, which will be described later.
(162) FIG. **18** is a partial sectional view of the printer **1** in a plane perpendicular to the right-left direction, in the state in which the peeling unit open button **52***b* is continuously pressed down to make the peeling roller cover **41** be at the open position and to make the peeling roller holder **42** be

at the protruding position. FIG. **18** does not show the thermal head **28** and the coil spring **55**, in order to make the shaft-receiving groove **25**, into which the shaft **28***a* of the thermal head **28** is inserted, clearly visible.
(163) As shown in FIG. **18**, the internal frame of the printer **1** is formed with the shaft-receiving groove **25** having a substantially L-shape. Although FIG. **18** shows only a shaft-receiving groove **25** that receives one of the pair of shafts **28***a* of the thermal head **28**, another shaft-receiving groove **25** that receives the other shaft **28***a* is also formed in the same manner.
(164) As shown by the enlarged drawing in FIG. **18**, the shaft-receiving groove **25** has a first groove **251** and a second groove **252**. Herein, each of positions P**1** and P**2** shows a position where the shaft **28***a* can be in the shaft-receiving groove **25**, in a virtual manner. In this disclosure, the state in which the shaft **28***a* is at the position P**1** may be referred to as a state in which the thermal head **28** is at the position P**1**; the state in which the shaft **28***a* is at the position P**2** may be referred to as a state in which the thermal head **28** is at the position P**2**.
(165) The first groove **251** extends in a direction in which the thermal head **28** moves to and away from the position P**1** (an example of a first position of a print head). The second groove **252** extends from the position P**1** to the position P**2** (an example of a second position of a print head) in a direction in which the coil spring **55** in front of the thermal head **28** biases the thermal head **28** (that is, in a rear direction; an example of a first direction). The shaft-receiving groove **25** is an L-shaped groove composed of the first groove **251** and the second groove **252**, and therefore, the position of the thermal head **28** can be switched between two positions P**1**, P**2** by this relatively simple shape. Herein, the position P**2** is a position at which the thermal head **28** cannot be removed by moving it upward, while the position P**1** is a position at which the thermal head **28** can be removed by moving it upward.
(166) The thermal head **28** is movable between the positions P**1** and P**2** in the direction of being biased by the coil spring **55**. Thus, in mounting the thermal head **28**, the thermal head **28** can be easily set to the position P**2** due to the biasing force of the coil spring **55**, simply by inserting the shaft **28***a* to the position P**1** along the first groove **251**.
(167) Next, a method of replacing the thermal head **28** will be described with reference to FIG. **19**.
(168) FIG. **19** illustrates a method of replacing the thermal head **28** and shows partial side views of a replacement-target thermal head **28** in states S**20** and S**21**.
(169) Normally, the replacement-target thermal head **28**, which is mounted to the printer **1**, is disposed at the position P**2** of the shaft-receiving groove **25**, as shown by the state S**20**. In this state, the whole thermal head **28** is biased to the platen roller **10** (not shown in FIG. **19**) (that is, in the rear direction) by the biasing force of the coil spring **55**, and the shaft **28***a* of the thermal head **28** is thereby stably positioned at the position P**2**.
(170) In order to remove the replacement-target thermal head **28**, it is moved from the position P**2** to the position P**1** in a direction opposite to a first direction, against the biasing force of the coil spring **55**, as shown by the state S**21**. The first direction is a direction in which the coil spring **55** biases the thermal head **28**, and the direction opposite to the first direction is a front direction. Subsequently, the replacement-target thermal head **28** is moved upward (an example of a second direction) from the position P**1**, and the shaft **28***a* of the replacement-target thermal head **28** is removed from the shaft-receiving groove **25**, whereby the replacement-target thermal head **28** is removed. At this time, the flexible cable **57** is connected to the connector **285** of the replacement-target thermal head **28** (refer to FIG. **21**B). Thus, the flexible cable **57** is disconnected from the connector **285** of the replacement-target thermal head **28**.
(171) After the replacement-target thermal head **28** is detached from the flexible cable **57**, a new thermal head **28** may be mounted to the printer **1** in a procedure reverse to the procedure of taking out the thermal head **28**.
(172) Specifically, the disconnected flexible cable **57** is first connected to the connector **285** of a new thermal head **28** (refer to FIG. **16**A). The new thermal head **28** is then inserted into the

position P**1** and is moved from the position P**1** to the position P**2** by the biasing force of the coil spring **55**. In more detail, the new thermal head **28** is moved downward (an example of an opposite direction to a second direction), and the shaft **28***a* of the new thermal head **28** is inserted into the shaft-receiving groove **25** from the first groove **251** (refer to FIG. **18**). At this time, insertion is performed while the end of the coil spring **55** (rear end of the coil spring is pressed forward (in the direction against the biasing force of the coil spring **55**) by the back surface **281***b* (surface facing forward of the printer **1**) of the new thermal head **28**. Upon reaching the position P**1**, the shaft **28***a* of the new thermal head **28** is moved to the position P**2** by the biasing force of the coil spring **55**, without requiring an operating force of an operator.

(173) Thus, the thermal head **28** is replaced as described above.

(174) The thermal head **28** is not disposed with the surface-mount devices, such as the connector, on the rear side (on the side disposed with the heat generating part **284**), as shown in FIG. **16**B, and it is thereby easy to replace. Also, in consideration of the coil spring **55** biasing the thermal head **28** rearward, if the thermal head **28** did not have a flat rear side, it would interfere with the internal frame on a rear side (e.g., a wall surface **21**; refer to FIG. **20**) and would be difficult to smoothly insert into the shaft-receiving groove **25**. In contrast, due to the thermal head **28** having a flat rear side, the new thermal head **28** can be smoothly inserted into the shaft-receiving groove **25**, although biased by the coil spring **55**.

(175) Mounting and removing of the thermal head **28** are performed when the peeling unit **4** is at the open position. In more detail, the peeling unit **4** at the closed position covers at least a part of the thermal head **28**, whereas the peeling unit **4** at the open position does not cover the thermal head **28**, as shown in FIG. **2**. In view of this, mounting and removing of the thermal head **28** are performed when the peeling unit **4** is at the open position.

(176) When the peeling unit **4** is at the closed position, none of other member is interposed between the peeling unit **4** and the thermal head **28**, and the peeling unit **4** directly covers at least a part of the thermal head **28**.

(177) With reference again to FIG. **18**, in the state in which the peeling roller cover **41** is at the open position (that is, the peeling unit **4** is at the open position), space for allowing mounting and removing the thermal head **28** having the shaft **28***a* at the position P**1**, is formed. Thus, an operator can remove the thermal head **28** from the printer **1** in accordance with merely the following operation process: opening the printer cover **3**; continuously pressing down the peeling unit open button **52***b* to make the peeling unit **4** be in the state shown in FIG. **18**; as described above, sliding the shaft **28***a* of the thermal head **28** from the position P**2** to the position P**1** against the biasing force of the coil spring **55**; and pulling up the thermal head **28**.

(178) In addition, in the printer **1** of this embodiment, at least a part of the rear side of the thermal head **28** is exposed to the paper roll-containing chamber **9**, as shown in FIG. **2**. With this structure, working space for taking out the thermal head **28** is ensured by temporarily removing the paper roll "R," which enables more easily taking out the thermal head **28**. Specifically, in sliding the shaft **28***a* of the thermal head **28** from the position P**2** to the position P**1**, an operator needs to apply an operating force to the thermal head **28** from a rear side to a front side, but the operating force is easily applied due to the space behind the thermal head **28**. Moreover, in pulling up the thermal head **28**, the space behind the thermal head **28** helps an operator in putting a hand therein and pulling up.

(179) In mounting the thermal head **28** to the printer **1**, an operation is performed in the order reverse to the operation in taking out the thermal head **28** from the printer **1**. As in the case described above, the peeling unit **4** is set to the state shown in FIG. **18**. Then, the shaft **28***a* of the thermal head **28** is inserted into the position P**1** from the first groove **251** of the shaft-receiving groove **25** while the end of the coil spring **55** (rear end of the coil spring **55**) is pressed forward (in the direction against the biasing force of the coil spring **55**) by the back surface **281***b* (surface facing forward of the printer **1**) of the thermal head **28**. The thermal head **28** is then moved to the

position P**2** by the biasing force of the coil spring **55**.
(180) Thus, the thermal head **28** can be easily replaced without using tools.
(181) In another embodiment, the shaft-receiving groove may have another shape, instead of the L-shape. The shaft-receiving groove may have, for example, a groove extending obliquely forward or extending obliquely rearward from the position P**1**, as long as the thermal head **28** can be attached and removed from the position P**1**. Alternatively, the shaft-receiving groove may have a U-shaped groove path between the positions P**1** and P**2** in such a manner that the position P**2** is provided at a position that the path reaches after starting from the position P**1** in FIG. **18**, extending forward, extending slightly downward, and then extending rearward, although this structure causes mounting and removing the thermal head **28** to be a little difficult. In this case, an operator can remove the shaft **28***a* of the thermal head **28** by moving it from the position P**2** to the position P**1** along the U-shaped groove.
Support Structure of Thermal Head **28**
(182) Next, a support structure of the thermal head **28** will be described with reference to FIGS. **20**, **21**A, and **21**B.
(183) First, a structure of the internal frame on a rear side of the thermal head **28** will be described with reference to FIG. **20**. FIG. **20** is a perspective view of a part of the internal frame along with components attached to the internal frame, a part of which is enlarged. FIG. **20** does not show the thermal head **28**.
(184) As shown in FIG. **20**, the internal frame has a wall surface **21** that is configured to face the rear surface of the thermal head **28**, behind an area to be disposed with the thermal head **28** (on a paper roll-containing chamber **9** side). The wall surface **21** is formed with a protrusion **211**. The protrusion **211** (an example of an abutting part) abuts on the rear surface of the thermal head **28** that is mounted. As shown in FIG. **20**, the abutting surface of the protrusion **211** is preferably curved so as to be convex toward the rear surface of the thermal head **28**.
(185) FIGS. **21**A and **21**B both illustrate forces that act on the thermal head **28** in the printer **1** of this embodiment; FIG. **21**A shows a cross section in a plane perpendicular to the upper-lower direction, and FIG. **21**B shows a cross section in a plane perpendicular to the right-left direction. FIGS. **21**A and **21**B have scales different from each other.
(186) As shown in FIG. **21**A, the protrusion **211** is provided at a position at which it abuts on a substantially center part in the right-left direction of the thermal head **28** that is mounted. In addition, the protrusion **211** is provided at a position at which it abuts on a substantially center position in the right-left direction between the pair of coil springs **55**, of the rear side of the thermal head **28**.
(187) The cutout **283***c* (refer to FIG. **16**B) is provided at the approximate center in the right-left direction of the thermal head **28**, as described above, and the protrusion **211** abuts on the thermal head **28** at the cutout **283***c*. The cutout **283***c* is not covered with the board **282** and exposes the heat dissipation plate **281** of the thermal head **28**, whereby the thermal head **28** is more stably supported.
(188) It should be noted that the cutout **283***c* is not necessarily provided. The protrusion **211** may support the thermal head **28** at an area of the board **282**, without the cutout **283***c* provided.
(189) The rear surface of the thermal head **28** is preferably provided with a recess having a shape corresponding to the protrusion **211**, at the position for abutting on the protrusion **211**. This causes the thermal head **28** to hardly deviate from the position for abutting on the protrusion **211** and to be more stably supported.
(190) In an embodiment, a recess may be provided in the wall surface **21** of the internal frame, whereas the rear surface of the thermal head **28** may be provided with a protrusion having a shape corresponding to the recess of the wall surface **21**. In this case, the thermal head **28** is also able to swing, but it is stably supported.
(191) The shape of the protrusion **211** shown in FIG. **20** is merely an example, and it can be

another shape that swingably supports the thermal head **28**. For example, the outer shape of the protrusion **211** may be a part of a spherical surface, instead of the shape shown in FIG. **20**.
(192) As shown in FIG. **21**A, in a plane view of the printer **1**, rearward restoring forces F**1** and F**2** of the pair of coil springs **55** act on the front side of the thermal head **28**, whereas a reaction force F**3** acts from the protrusion **211** abutting on the rear side of the thermal head **28**. Herein, the protrusion **211** is at the approximate center position in a top view of the printer **1**, and thus, the thermal head **28** is able to swing around a fulcrum at the protrusion **211**, in a clockwise direction and a counterclockwise direction in FIG. **21**A.
(193) As shown in FIG. **21**B, in a side view of the printer **1**, the rearward restoring forces F**1** and F**2** (restoring force F**2** is not visible in FIG. **21**B) of the pair of coil springs **55** act on the front side of the thermal head **28**. A reaction force F**4** from the platen roller **10** acts on the rear side of the thermal head **28**, above the points of application of the restoring forces F**1** and F**2**. The reaction force F**3** from the protrusion **211** acts on the rear side of the thermal head **28**, under the points of application of the restoring forces F**1** and F**2**. Thus, the thermal head **28** is able to swing around a fulcrum at the protrusion **211**, in a clockwise direction and a counterclockwise direction in FIG. **21**B.
(194) In addition, in a side view of the printer **1**, the points of applying the biasing forces of the coil springs **55** to the thermal head **28** are between the position at which the thermal head **28** receives the reaction force from the platen roller **10** and the position at which the protrusion **211** supports the rear side of the thermal head **28**. With this structure, the biasing forces of the coil springs **55** are received at an upper part and a lower part, whereby the thermal head **28** is supported with good balance.
(195) In FIG. **21**B, the surface on which the protrusion **211** abuts (that is, the surface on which the heat dissipation plate **281** is exposed by the cutout **283***c*), and the surface corresponding to the heat generating part **284**, are preferably in the same reference plane on the rear side of the thermal head **28**. This enables pressing the heating elements of the thermal head **28** against the platen roller **10** at an appropriate angle.
(196) As shown in FIGS. **21**A and **21**B, the thermal head **28** is able to swing around a fulcrum at the protrusion **211** in a clockwise swinging direction and a counterclockwise swinging direction in a side view of the printer **1**. The thermal head **28** is also able to swing around a fulcrum at the protrusion **211** in a clockwise swinging direction and a counterclockwise swinging direction in a plane view of the printer **1**. Thus, the thermal head **28** uniformly applies pressure to the platen roller **10** in printing. The reason of this is as follows.
(197) In a printer having an existing thermal head, the thermal head is fixed, for example, at two points, by using screws, shafts, brackets, or the like, so as to be mounted to an internal frame or a housing of the printer. In such a case, due to deviation of the mounted position, the pressure of the thermal head abutting on a platen roller may not be uniform in an axial direction of the platen roller, which may cause degradation in print quality.
(198) On the other hand, in this embodiment, the thermal head **28** is able to swing around a fulcrum at the protrusion **211** in a side view and in a plane view of the printer **1**. With this structure, the thermal head **28** can follow and maintain uniform pressure on the platen roller **10**, for example, even when there is a mounting error of the platen roller **10**, circular runout of the platen roller **10** is large in rotating, or a rugged surface label is temporarily attached on a liner.
(199) Moreover, the thermal head **28** is movable between the positions P**1** and P**2** (refer to FIG. **18**) in the direction of being biased by the coil spring **55**, and thus, the thermal head **28** is not prevented from swinging around a fulcrum at the protrusion **211**.
(200) In some printers having an existing thermal head, a fulcrum shaft is provided at a lower part of the thermal head, and this shaft is fixed to a printer body to enable the thermal head to swing in a side view. However, unlike the printer **1**, this thermal head cannot be replaced without using tools. On the other hand, the printer **1** is superior to existing ones in that the thermal head **28** can be

replaced without using tools while enabling to swing in a side view and in a plane view of the printer **1**.
(201) In another embodiment, protrusions **211** may be provided at two positions separated in the right-left direction on the wall surface **21** shown in FIG. **20**. Also in this case, the thermal head **28** is able to swing around fulcrums at the protrusions **211** in a side view of the printer **1**. Even in the case in which the thermal head **28** is able to swing only in a side view of the printer **1**, degradation in print quality is prevented.
(202) In another embodiment, protrusions **211** may be provided at two positions separated in the upper-lower direction on the wall surface **21** shown in FIG. **20**. Also in this case, the thermal head **28** is able to swing around fulcrums at the protrusions **211** in a plane view of the printer **1**. Even in the case in which the thermal head **28** is able to swing only in a plane view of the printer **1**, degradation in print quality is prevented.
(203) As shown in FIG. **16**A, the flexible cable **57** is detachably connected to the thermal head **28**. The flexible cable **57** is connected from the connector **285** of the thermal head **28** that is mounted to the printer **1**, to the circuit board (not shown) at a front part of the printer **1**, as shown in FIG. **21**B. The flexible cable **57** is fixed at a fixing position **24***a* on an upper surface of a bracket **24** in front of the thermal head **28**, for example, by screwing or adhesive.
(204) A cable-containing chamber **59** for containing the flexible cable **57** is formed between the thermal head **28** and the circuit board. The cable-containing chamber **59** is configured to contain the relatively long flexible cable **57** between the connector of the thermal head **28** and the fixing position **24***a*. With this structure, when removed, the thermal head **28** can be moved to a position sufficiently higher than the printer **1** based on the fixed position **24***a*. This makes it easy to remove the flexible cable **57** from the connector of the thermal head **28** and to replace with a new thermal head **28**.
(205) However, the cable-containing chamber **59** is not necessarily formed. Even in this case, although the cable length from the connector **285** of the thermal head **28** to the fixing position **24***a* is reduced, it is possible to remove the flexible cable **57** from the connector **285** and to replace the thermal head **28**.
(206) As shown in FIGS. **21**A and **21**B, the cable-containing chamber **59** is formed in space between the platen-holding bracket **27** and the thermal head **28**. Thus, the space that is formed by the platen-holding bracket **27** having a U-shape in a plane view is efficiently used.
(207) The cable-containing chamber **59** may not be formed as the space between the platen-holding bracket **27** and the thermal head **28**. In one example, the flexible cable **57** extending from the connector of the thermal head **28** may be passed under the platen-holding bracket **27**, and a containing chamber may be provided on a front side of the platen-holding bracket **27**.
(208) As described above, in the printer **1**, the surface-mount devices are not mounted on the rear surface of the thermal head **28** and are thereby protected from water, etc., which may enter from the ejection part **20**.
(209) In the printer **1**, space for allowing mounting and removing the thermal head **28** is formed when the peeling unit **4** is at the open position not covering the thermal head **28**, which improves the workability in replacing the thermal head **28**. Moreover, the thermal head **28** is biased rearward (in a direction to the platen roller **10**) and is movable along this direction between the first position for allowing mounting and removing the thermal head **28** and the second position for restricting mounting and removing of the thermal head **28**. Thus, the thermal head **28** can be removed only by moving it from the second position to the first position, and tools and the like are not necessary.
(210) In the printer **1**, the thermal head **28** is able to swing around a fulcrum at the protrusion **211** in a clockwise swinging direction and a counterclockwise swinging direction in a side view of the printer **1**, and the thermal head **28** is also able to swing around a fulcrum at the protrusion **211** in a clockwise swinging direction and a counterclockwise swinging direction in a plane view of the printer **1**. Thus, the thermal head **28** uniformly applies pressure to the platen roller **10** in printing,

and it is possible to prevent degradation in print quality due to the method of mounting the thermal head.

Another Embodiment of Thermal Head

(211) Next, a thermal head **28**A according to another embodiment will be described with reference to FIGS. **22**A to **25**.

(212) FIG. **22**A is a perspective front view of the thermal head **28**A, and FIG. **22**B is a perspective rear view of the thermal head **28**A. FIG. **23** is a perspective view of a plate member included in the thermal head **28**A. FIG. **24** is a perspective view of the thermal head **28**A, as seen from a viewpoint different from those of FIGS. **22**A and **22**B.

(213) It is clear from a comparison between FIGS. **22**A and **22**B and FIGS. **16**A and **16**B that the thermal head **28**A differs from the thermal head **28** in having a plate member **7**.

(214) The plate member **7**, which is a member formed of a metal material such as stainless steel, is fastened to the heat dissipation plate **281** with screws. As shown in FIG. **23**, the plate member **7** has a base **71**, projecting pieces **72**L and **72**R, and a projecting plate **73**.

(215) The projecting pieces **72**L and **72**R project from both ends of the base **71** in a direction perpendicular to a main surface of the base **71** (that is, in a direction perpendicular to the surface **281***a* when they are attached to the heat dissipation plate **281**). In the state in which the plate member **7** is attached to the heat dissipation plate **281**, the projecting pieces **72**L and **72**R project on a side mounted with the heat generating part **284**, as shown in FIG. **24**. The projecting pieces **72**L and **72**R have edge parts **721**L and **721**R at ends.

(216) The projecting piece **72**L is formed with a hole **72***a*, whereas the projecting piece **72**R is formed with a U-shaped groove **72***b*. As shown in FIGS. **22**A and **22**B, one of the pair of shafts **28***a* is inserted in the hole **72***a*, and the other shaft **28***a* is inserted in the U-shaped groove **72***b*. One of the edge parts **721**L and **721**R is formed with a hole, and the other is formed with a U-shaped groove. This facilitates attaching the plate member **7** to the heat dissipation plate **281**.

(217) In the state in which the plate member **7** is attached to the heat dissipation plate **281**, the projecting plate **73** projects on a side mounted with the relatively tall surface-mount devices (e.g., the connector **285**, the EEPROM **286**, and the diode **287**), as shown in FIG. **22**A.

(218) The projecting plate **73** is provided between the projecting pieces **72**L and **72**R over the longitudinal direction of the base **71** and projects from the base **71** in a direction opposite to the projecting pieces **72**L and **72**R.

(219) The base **71** is formed with two holes **71***a* for allowing screws to pass in mounting the plate member **7** to the heat dissipation plate **281**. The base **71** has two projections **711**. As shown in FIG. **22**A, the projections **711** are disposed so as to not interfere with the surface-mount devices when the plate member **7** is attached to the heat dissipation plate **281**.

(220) Hereinafter, effects of the thermal head **28**A having the plate member **7** will be described with reference to FIG. **25**. FIG. **25** is a side view showing a positional relationship between the thermal head **28**A and the platen-holding bracket **27**.

(221) As described above, when the printer cover **3** is at the closed position, the platen shaft **10***a* of the platen roller **10**, which is attached to the printer cover **3**, is fitted in the groove **27***b* of the platen-holding bracket **27**, whereby the printer cover **3** is held. In a case of the thermal head **28** that does not have the plate member **7**, when an operator presses the printer cover **3** from above, for closing the printer cover **3** for example, the platen roller **10** may deviate downward from a designed position at which the platen roller **10** and the thermal head **28** abut on each other. This causes variations in density of printing. Further, the thermal head **28** is fitted to the shaft-receiving groove **25** (refer to FIG. **18**), which is provided in the internal frame. The one end of the shaft **27***a* of the platen-holding bracket **27** is inserted in the boss **52***a* of the peeling unit open lever **52**, whereas the other end of the shaft **27***a* is inserted in the boss provided to the internal frame (refer to FIG. **5**). Thus, the position at which the platen roller **10** and the thermal head **28** abut on each other is susceptible to accumulated errors in assembling components and tends to deviate from the designed

position.

(222) The drawback of the thermal head **28** noted above is overcome by the thermal head **28**A.

(223) As shown by an enlarged drawing in FIG. **25**, in the case in which the thermal head **28**A is mounted to the printer **1**, instead of the thermal head **28**, upper ends of the edge parts **721**L and **721**R of the plate member **7** of the thermal head **28**A are disposed at positions higher than rims that form the grooves **27***b* of the platen-holding bracket **27**. Thus, the platen shaft **10***a* that is fitted in the platen-holding bracket **27** is in contact with the edge parts **721**L and **721**R in the grooves **27***b*. This makes it difficult for the platen roller **10** to deviate downward from the designed position at which the platen roller **10** and the thermal head **28** abut on each other, even when an operator presses the printer cover **3** from above. This is because the plate member **7** is integrally coupled to the heat dissipation plate **281** mounted with the heat generating part **284**, whereby a relative positional relationship between the platen roller **10** and the heat generating part **284** is unlikely to be affected even when the platen shaft **10***a* presses down the edge parts **721**L and **721**R.

(224) With reference again to FIG. **25**, in the state in which the thermal head **28**A is mounted to the printer **1**, the projecting plate **73** of the plate member **7** projects toward the front side of the printer **1**. Thus, an upper part of the cable-containing chamber **59**, which is formed in front of the thermal head **28**A, is covered with the projecting plate **73**. This structure prevents dust from entering the printer **1** from the outside, resulting in preventing dust from adhering to upper surface portions of the surface-mount devices disposed on the front side of the thermal head **28**A. That is, the projecting plate **73** functions as a hood. In particular, as shown in FIG. **2**, replacement of the paper roll “R” is performed while the printer cover **3** is maintained at the open position, and dust tends to enter the printer **1**. However, in this situation, it is also possible to protect the surface-mount devices of the thermal head **28**A from dust.

(225) From another point of view, providing the projecting plate **73** improves strength of the plate member **7**.

(226) Although some embodiments of the printer of the present invention are described above, the present invention should not be limited to the foregoing embodiments. In addition, the embodiments described above can be variously modified and altered within the scope not departing from the gist of the present invention. For example, respective technical features described in the foregoing embodiments can be combined with one another as appropriate, unless technical contradiction occurs.

(227) For example, the structures and the mounting and removing methods of the thermal heads **28** and **28**A are not technically related to the structure of the peeling unit **4** and the method of switching the issue modes, and therefore, they may be employed in a printer without the peeling unit **4**. Conversely, the structure of the peeling unit **4** and the method of switching the issue modes may be employed in a printer that uses a structure and an mounting and removing method of a thermal head different from those of the thermal heads **28** and **28**A.

(228) A case in which some parts (e.g., shafts and ends of springs) of components inside the printer **1** are coupled to the internal frame is described here; but the structure is not limited thereto, and these parts may be coupled to the body case **2**.

(229) Although a case of using a print medium that is a continuous paper having a plurality of labels temporarily attached on a liner is described in the foregoing embodiments, the print medium is not limited thereto. For continuous issuing or for a printer not provided with a peeling unit, for example, a continuous label having an adhesive surface on one side (label without a liner), a continuous sheet without an adhesive surface (continuous sheet), or a material other than papers such as a film, which is printable by a thermal head, may also be used as a print medium. In addition, in a case of feeding a label having an exposed adhesive due to no liner, a feeding path may be coated with a non-adhesive material, and a non-adhesive roller containing silicone or the like may be provided as a platen roller.

(230) The present invention is related to Japanese Patent Application No. 2020-191576 filed with

the Japan Patent Office on Nov. 18, 2020, the entire contents of which are incorporated into this specification by reference.

Claims

1. A printer comprising: a feed roller configured to feed a print medium; a print head configured to pinch the print medium with the feed roller and to print on the print medium; at least a biasing member configured to bias the print head in a first direction toward the feed roller; and an abutting part that abuts on the print head, wherein: the print head has a first side biased by the biasing member, and a second side opposite the first side, the second side facing the feed roller, wherein the print head comprises: a heat generating part located on the second side of the print head, and a protective layer located on the second side of the print head, the protective layer extends from the heat generating part to a region of the print head without the protective layer, at which the abutting part abuts on the print head, in a side view of the printer, a point at which a biasing force of the biasing member is applied to the print head is between a position at which the print head receives a reaction force from the feed roller and a position at which the abutting part supports the second side of the print head, and the print head is swingable around a fulcrum at the abutting part in a clockwise swinging direction and a counterclockwise swinging direction in a side view of the printer.
2. The printer according to claim 1, wherein: the at least a biasing member comprises a pair of biasing members arranged in a width direction of the printer, the abutting part abuts on the second side of the print head, at a substantially center position of the pair of biasing members, and the print head is swingable around a fulcrum at the abutting part in a clockwise swinging direction and a counterclockwise swinging direction in a plan view of the printer.
3. The printer according to claim 2, wherein the second side of the print head is provided with a recess or a protrusion having a shape corresponding to the abutting part, at a position for abutting on the abutting part.
4. The printer according to claim 2, wherein the print head is movable in the first direction between a first position and a second position, the first position being a position at which mounting and removing the print head are enabled, the second position being a position at which mounting and removing the print head are restricted.
5. The printer according to claim 4, wherein: the print head comprises: a base on which the heat generating part is mounted, and a pair of shafts that extend outward from both ends of the base, and the printer further comprises a pair of grooves that receives the pair of shafts, the pair of grooves including a first groove and a second groove, the first groove extending from the first position in a direction for mounting and removing the print head, the second groove extending in the first direction from the first position to the second position.
6. The printer according to claim 1, wherein the second side of the print head is provided with a recess or a protrusion having a shape corresponding to the abutting part, at a position for abutting on the abutting part.
7. The printer according to claim 6, wherein the print head is movable in the first direction between a first position and a second position, the first position being a position at which mounting and removing the print head are enabled, the second position being a position at which mounting and removing the print head are restricted.
8. The printer according to claim 7, wherein; the print head comprises: a base on which the heat generating part is mounted, and a pair of shafts that extend outward from both ends of the base, and the printer further comprises a pair of grooves that receives the pair of shafts, the pair of grooves including a first groove and a second groove, the first groove extending from the first position in a direction for mounting and removing the print head, the second groove extending in the first direction from the first position to the second position.

9. The printer according to claim 1, wherein the print head is movable in the first direction between a first position and a second position, the first position being a position at which mounting and removing the print head are enabled, the second position being a position at which mounting and removing the print head are restricted.
10. The printer according to claim 9, wherein: the print head comprises: a base on which the heat generating part is mounted, and a pair of shafts that extend outward from both ends of the base, and the printer further comprises a pair of grooves that receives the pair of shafts, the pair of grooves including a first groove and a second groove, the first groove extending from the first position in a direction for mounting and removing the print head, the second groove extending in the first direction from the first position to the second position.
11. The printer according to claim 1, wherein: a surface on which the abutting part abuts, and a surface on which the heat generating part is formed, are in the same reference plane on the second side of the print head.
12. The printer according to claim 1, wherein the second side of the print head is provided with a recess or a protrusion having a shape corresponding to the abutting part, at a position for abutting on the abutting part.
13. The printer according to claim 1, wherein the print head is movable in the first direction between a first position and a second position, the first position being a position at which mounting and removing the print head are enabled, the second position being a position at which mounting and removing the print head are restricted.
14. The printer according to claim 13, wherein; the print head comprises: a base on which the heat generating part is mounted, and a pair of shafts that extend outward from both ends of the base, and the printer further comprises a pair of grooves that receives the pair of shafts, the pair of grooves including a first groove and a second groove, the first groove extending from the first position in a direction for mounting and removing the print head, the second groove extending in the first direction from the first position to the second position.
15. The printer according to claim 1, wherein the protective layer is a ceramic board.
16. The printer according to claim 1, wherein the protective layer is a coating layer.
17. The printer according to claim 1, wherein the protective layer extends from the second side of the print head to the first side of the print head.
18. A printer comprising: a feed roller configured to feed a print medium; a print head configured to pinch the print medium with the feed roller and to print on the print medium; at least a biasing member configured to bias the print head in a first direction toward the feed roller; and an abutting part that abuts on the print head, wherein: the print head has a first side biased by the biasing member, and a second side opposite the first side, the second side facing the feed roller, wherein the print head comprises: a base on which heating elements are mounted, and a pair of shafts that extend outward from both ends of the base, in a side view of the printer, a point at which a biasing force of the biasing member is applied to the print head is between a position at which the print head receives a reaction force from the feed roller and a position at which the abutting part supports the second side of the print head, the print head is swingable around a fulcrum at the abutting part in a clockwise swinging direction and a counterclockwise swinging direction in a side view of the printer, the print head is movable in the first direction between a first position and a second position, the first position being a position at which mounting and removing the print head are enabled, the second position being a position at which mounting and removing the print head are restricted, and p**1** the printer further comprises a pair of grooves that receives the pair of shafts, the pair of grooves including a first groove and a second groove, the first groove extending from the first position in a direction for mounting and removing the print head, the second groove extending in the first direction from the first position to the second position.